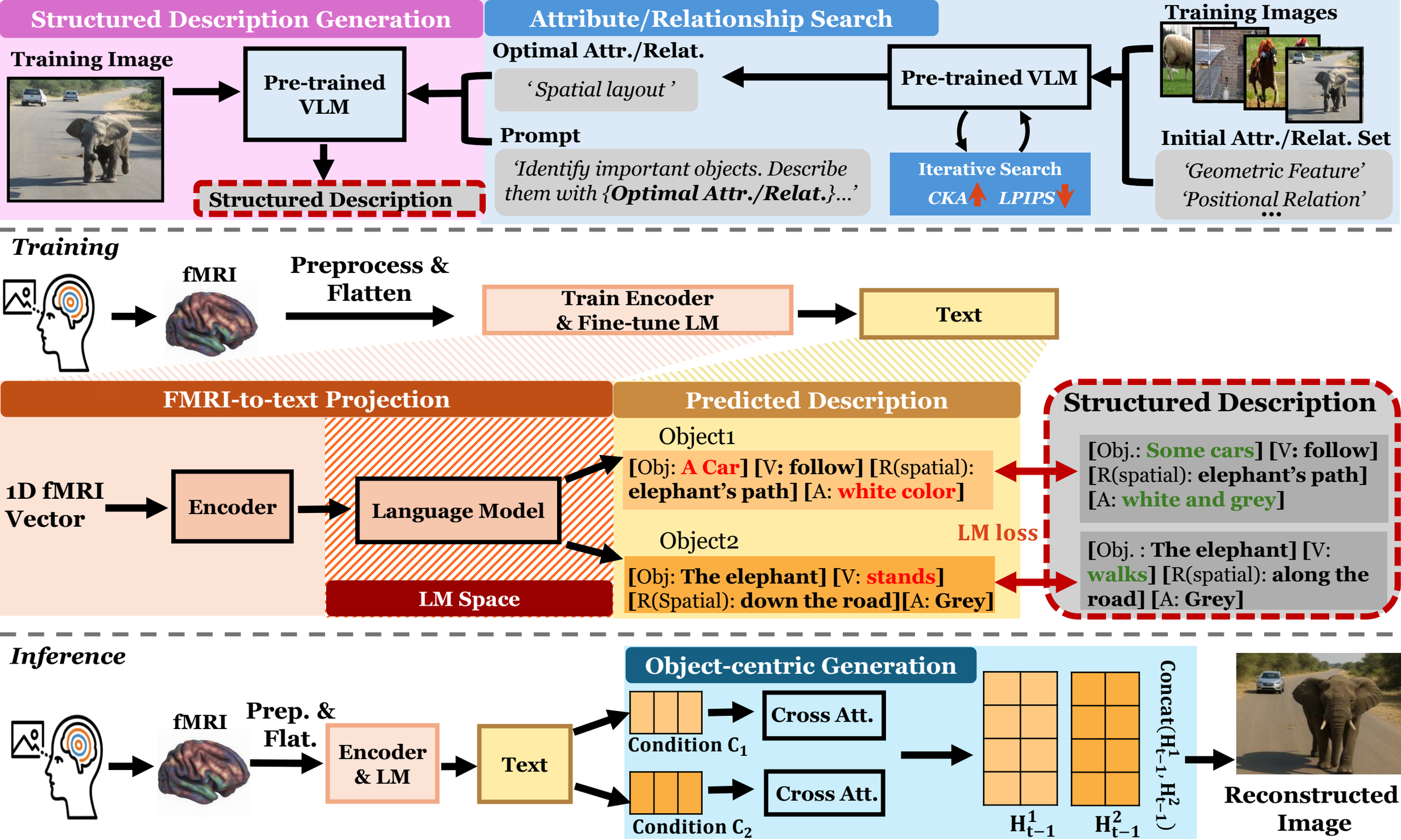

Structured Description Generation
Attribute/Relationship Search
Training Image
Pre-trained VLM
Structured Description
Optimal Attr./Relat.
'Spatial layout'
Prompt
'Identify important objects. Describe them with {Optimal Attr./Relat.}...'
Pre-trained VLM
Iterative Search
CKA
LPIPS
Training Images
Initial Attr./Relat. Set
'Geometric Feature'
'Positional Relation'
...
Training
fMRI
Preprocess & Flatten
Train Encoder & Fine-tune LM
Text
FMRI-to-text Projection
Predicted Description
1D fMRI Vector
Encoder
Language Model
LM Space
Object1
[Obj: A Car] [V: follow] [R(spatial): elephant's path] [A: white color]
LM loss
Object2
[Obj: The elephant] [V: stands] [R(Spatial): down the road][A: Grey]
Structured Description
[Obj.: Some cars] [V: follow] [R(spatial): elephant's path] [A: white and grey]
[Obj. : The elephant] [V: walks] [R(spatial): along the road] [A: Grey]
Inference
fMRI
Prep. & Flat.
Encoder & LM
Text
Object-centric Generation
Condition C_1
Cross Att.
Condition C_2
Cross Att.
H^1_{t-1}
H^2_{t-1}
Concat(H^1_{t-1}, H^2_{t-1})
Reconstructed Image